MIL-C-43227A

11 May 1973
SUPERSEDING
MIL-C-43227
28 April 1964

MILITARY SPECIFICATION

CUTTERS, PAPER, GUILLOTINE, HAND OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers manually operated paper cutters designed to cut quantities of paper simultaneously to a pre-determined size by a guillotine action of the cutting blade.

1.2 Classification. The paper cutters shall be of the following types and sizes as specified (see 6.2):

Type I - Bench model.
Size 1 - 19 inch.
Size 2 - 23 inch.

Type II - Floor model.
Size 1 - 26-1/2 inch.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-3684 - Printing, Duplicating and Book Binding Equipment, Packaging of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 3610

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MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3, 6.2, and 6.3).

* 3.2 Standard product. The paper cutters delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof shall be new and interchangeable.

* 3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial paper cutters provided that the completed item complies with all provisions of this specification.

3.4 Design. The paper cutters shall be designed to cut paper stock by means of a hand lever actuated cutting knife descending across the paper stock in a slicing motion. The cut shall be parallel with a movable back gage and perpendicular to fixed right and left hand paper guides. The handlever on type II cutters shall be counter balanced for positive return after each cut.

3.5 Detail requirements.

3.5.1 Capacities and dimension. The paper cutters shall have capacities and dimensions in accordance with table I.

TABLE I. Capacities and dimensions (inches)

	Type I		Type II
	Size 1	Size 2	Size 1
Cutting width (throat width) (min.)	19	23	26-1/2
Shall cut pile of paper	2-1/2	2-1/2	3-1/4
Minimum cut, back of knife + 0 - 1/4	3/4	3/4	3/4

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TABLE I. Capacities and dimensions (inches) (cont'd)

	Type I		Type II
	Size 1	Size 2	Size 1
Table space, front of knife (min.)	11	12-1/4	13-1/2
Length of cut, back of knife + 1/2 - 0	24	24	26-1/2
Height of table from base ± 1	5-1/2	5-1/2	34
Width, overall (with lever raised) (max.)	40	44	54
Length, overall (maximum)	46	46	60
Height, overall (with lever raised) (max.)	44	44	70

3.5.2 Supporting structure. The supporting structure shall be so designed and constructed as to support the assembled paper cutters in such a manner as to meet the performance requirements specified herein. Mounting holes, not less than 3/8 inch diameter, for type I cutters and 9/16 inch diameter for type II cutters, shall be located adjacent to the four corners of the base to permit bolting the assembly to a supporting surface.

* 3.5.3 Table, cutting. The cutting table shall have a smooth top surface uniformly finished flat within plus or minus 0.003 inch on the width between right and left hand side gages. Flat within plus or minus .010 on the length of the table behind the cutting stick when tested in accordance with 4.5.1.

3.5.3.1 Rules.

3.5.3.1.1 Type I paper cutters. Type I paper cutters shall be provided with metal rules inlaid in the cutting table. The rules shall run perpendicular to the cutting knife edge along the full length of the table in front of the cutting knife and back in full length of usable table capacity shown in table I. Rules shall be graduated in sixteenths of an inch, numbered every inch, with graduations and number characters contrasting with the immediate background.

* 3.5.3.1.2 Type II paper cutters. Type II paper cutters shall be provided with a front rule as specified in 3.5.3.1.1. The rule back of the cutting knife shall be a movable back gage measuring tape, with position indicator, coordinated with the movement of the back gage (see 3.5.3.3.2). In addition to the 1/16 inch graduations, both rules for type II paper cutters shall be graduated to 1/3, 1/6 and 1/12 inch. The back gage measuring tape may be additionally graduated in millimeters at the manufacturers option.

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3.5.3.2 Cutting sticks. Each paper cutter shall be provided with not less than 4 close grain hardwood or plastic cutting sticks. The cutting sticks shall be free from bows, twists, or splits, and shall be sized for a push fit in the table top cutting stick recess. The recess shall be slightly off center from the cutting knife edge to extend the serviceable life of the cutting stick.

3.5.3.3 Gages.

3.5.3.3.1 Side gages. Right and left hand side gages shall be fixed to the table top and spaced to comply with the applicable throat width specified in table I. The working plane of the side gages shall be square with the table top and the cutting knife edge. Both side gages shall extend not less than 1/2 the working depth of the table top in back of the cutting knife and not less than 1/4 the working depth in front of the cutting knife. The gage height shall be not less than the applicable pile height specified in table I.

3.5.3.3.2 Back gage. A movable fingered back gage shall be provided on the table top, back of the cutting knife. The back gage shall be movable throughout the range of distances specified in table I. Throughout this range, the working plane of the back gage shall be parallel to the cutting knife edge and square with the table top. Adjusting means shall be provided as necessary to maintain squareness and parallelism, as applicable, between the working plane of the back gage and knife edge, and the table top and side gages. Movement of the back gage shall be controlled by a handwheel and locking device at the front of the machine.

3.5.3.3.2.1 Type II back gage option. When specified (see 6.2), type II back gage shall be a three piece adjustable split type joined by nuts and bolts to permit shearing three different paper lengths simultaneously, to predetermined settings.

3.5.4 Knife bar and cutting knife.

3.5.4.1 Knife bar and cutting knife. A knife bar shall be incorporated in the upper frame assembly to retain the cutting knife. The cutting motion of the knife bar shall be accomplished through means of a hand lever interconnected to the knife bar and frame assembly. The knife bar shall have a machined cutting knife retaining seat. Adjusting devices shall be incorporated to permit adjusting the cutting knife edge to the cutting stick, when the bottom of the cutting stroke has been reached, and to compensate for wear in the cutting bar slides.

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* 3.5.4.2 Cutting knife. Two cutting knives shall be furnished with each paper cutter. The cutting knives shall be of hardened and ground steel so designed and constructed to prevent drag on the paper pile during the cutting operation, and to provide for solid seating and bolting to the knife bar. The leading or cutting edge of the cutting knife shall be taper ground and honed to minimize resistance during the cutting operation and to produce clean sharp cut edges. Knife edge guards shall be furnished with each cutting knife for use in removing and replacing the cutting knife in the knife bar.

3.5.4.3 Safety lock. Each paper cutter shall be equipped with a positive acting safety lock to hold the knife bar in the "up" position at the conclusion of each cutting stroke. Safety lock release shall be such as to require both hands of the operator either on the operating lever or the left hand on the safety lock release and the right hand on the operating lever.

3.5.5 Paper clamp. Each paper cutter shall be provided with a fingered paper clamp arranged to mesh with the fingered back gage for close up work. The paper clamp shall be equipped with a detachable clamping plate to increase surface area of the fingered clamp. The paper clamp shall be lowered and raised by a hand wheel. The clamping action shall exert sufficient pressure to hold the paper pile firmly to insure accurate cutting.

* 3.6 Performance. When tested as specified in 4.5.2.2 the sheets in a trimmed pile shall not vary in trimmed dimension more than 0.015 inch; the actual trimmed dimension shall correspond with the rule readings within 0.010 inch and adjacent edges of folded sheets shall be parallel within 0.010 inch. The trimmed edge of the piles shall be smooth and individual sheets shall have straight and clean edges.

3.7 Floor stand. When specified (see 6.2), type I paper cutters shall be furnished with an all steel floor stand of a height to position the cutting table surface 29 to 33 inches above the floor. The top and three sides shall be fully inclosed. At least one shelf shall be incorporated 8 to 12 inches above the floor running full width and depth of the inside dimensions. Mounting holes, corresponding to those in the paper cutter base, shall be located in the floor stand top plate. All exposed corners and edges of the floor stand shall be box flanged or rounded, for strength and appearance.

3.8 Finish. Surface preparation and finish shall be in accordance with the manufacturer's regular commercial practice. All painted surfaces shall be uniformly coated and free from runs, wrinkles, orange peel, teardrops, grit, sags or areas of thin or no film.

3.9 Marking for identification. Marking for identification shall be in accordance with the unit marking requirements of MIL-STD-130.

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* 3.10 Tools. Each paper cutter shall be furnished with a set of tools required for adjustment and maintenance of the unit furnished.

* 3.11 Workmanship. The items furnished shall be free from defects which may cause injury to personnel or damage to the machine.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required, the cutter shall be examined for defects as specified in table II, for dimensions specified, and tested as specified in 4.5.1 thru 4.5.2.3. The presence of any visual defects and any dimensions not as specified or failure of test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.4.2 End item inspection. The inspection lot shall be all cutters of the same type and size offered for inspection at one time.

4.4.2.1 Visual inspection. The paper cutter shall be examined for the defects in table II. The inspection level shall be II with an AQL (acceptable qaquality level) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE II. Visual defects, end item

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's regular commercial finish		X
	Blistered, checked, or peeling		X
	Evidence of rust	X	
	Tacky (not dry to touch)		X
	Foreign matter imbedded		X
	Painted finish not uniformly coated or free from runs, wrinkles, orange peel, teardrops, grit, sags, or areas of thin or no film		X
	Component or assembly that is painted or otherwise coated, affecting serviceability		X
Design	Not specified type or size	X	
	Any characteristic not in accordance with specified requirements	X	
Construction:			
Supporting structure	Mounting holes not provided in base	X	
	Mounting holes not located as specified		X
	Mounting holes smaller than specified		X
Cutting table rules	Not specified type		X
	Not mounted as specified		X
	Graduations not as specified	X	
	Graduations and number characters not in contrast with background		X
	Measuring tape position indicator missing (type II)	X	
Cutting sticks	Not furnished	X	
	Less than 4 furnished		X
	Not close grain hardwood or plastic	X	
	Bow, split or twist	X	
	Will not enter table top recess	X	
Cutting stick recess	Cutting stick recess not provided in table top	X	
	Recess not located under cutting knife	X	
	Recess not offset to permit rotation or reversal of cutting stick		X

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TABLE II. Visual defects, end item (cont'd)

Examine	Defect	Classification Major	Minor
Construction: (cont'd)			
Side gages	Working plane not square with cutting table (check with toolmakers square)		X
Back gage	Not fingered to mesh with paper clamp	X	
	Adjusting means for squareness and parallelism not provided		X
	Not controlled by handwheel	X	
	Locking device missing		X
	Not 3 piece adjustable type (when required)	X	
Knife bar	Does not descend in slicing motion	X	
	Handle not counterbalanced (applicable to type II cutters)	X	
	Cutting knife retaining seat not machined		X
	Adjusting means for wear or adjustment to cutting stick not provided	X	
Cutting knife	Cutting edge not taper ground and honed		X
	Less than two furnished	X	
	Does not seat solidly in knife bar	X	
	Knife lifter missing	X	
	Knife lifters not positive engaging	X	
	Less than two knife guards	X	
Paper clamp	Not fingered to mesh with back gage	X	
	Detachable clamping plate not furnished		X
	Not handwheel activated		X
Floor stand (when applicable)	Not inclosed on three sides		X
	Shelf not provided or not full depth or width	X	
	Mounting holes not provided or holes do not mate with holes in paper cutting base	X	
	Exposed edges and corners not box flanged or rounded		X
Tools	Adjustment, maintenance and repair tools not furnished		X

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TABLE II. Visual defects, end item (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship	Any component missing or not as specified	X	
	Component fractured; split, punctured, dented, bowed, malformed, or otherwise impaired	X	
	Component misplaced, loose or not in proper alignment	X	
	Sharp burr, sliver, or splinter	X	
	Operation omitted or not properly performed	X	
	Adjustable assembly that cannot be adjusted to perform intended function	X	
	Functioning component inoperable	X	
	Functioning component that requires abnormal force to operate		X
	Component not accessible for servicing, when required		X
Marking for identification	Missing, incomplete, not legible, not affixed in prescribed manner		X

* 4.4.2.2 Dimensional examination. The paper cutter shall be examined for compliance with dimensions and weight specified. Any dimension or weight not within this specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

* 4.4.2.3 End item testing. When a first article is not required, the first produced paper cutter shall be examined and tested as specified in 4.5.1 thru 4.5.2.3. Each paper cutter or each sample paper cutter shall be tested as specified in 4.5.1 thru 4.5.2.3. Failure to pass a test shall be cause for rejection of the inspection lot. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed as defects per hundred units.

4.4.3 Examination of preparation for delivery. Examination of preservation, packaging, packing and marking shall be in compliance with section 5 and specification MIL-P-3684.

* 4.5 Test methods.

4.5.1 Table, top surface test. Table top surface shall be checked by placing a machinists straightedge, not less than 18 inches long, at various

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positions parallel to the edges of the table top, and along the diagonals, and testing with a feeler gage to determine compliance with 3.5.3.

4.5.2 Performance tests.

4.5.2.1 Material. The supplier shall furnish paper for determining cutting accuracy. The paper shall be sulphite bond or writing, substance 16, or book paper substance 50, 60 or 70, in sheet size and pile height as indicated in table III.

TABLE III. Sheet sizes and pile height

Paper cutter	Sheet size (inches)	Pile height (inches)
Type I, size 1	19 by 26	2-1/2
Type I, size 2	23 by 26	2-1/2
Type II, size 1	26 by 30	3-1/4

* 4.5.2.2 Cutting accuracy. The applicable paper pile shall be placed in the machine and cut to the maximum size, not less than 2 intermediate sizes and the minimum size for the machine under test. Advancement of the back gage during testing shall be with the paper pile located in the machine. Placement of the paper pile shall be against the back gage and alternately against the left and right side gages. Measurements shall be made of the top and bottom 6 sheets and random intermediate sheets of each cut pile to determine compliance with the actual paper size, uniformity of size and rule reading requirements in 3.6. In addition, the measured sheets in each cut pile shall be folded with the opposing edges carefully aligned to determine parallelism as required in 3.6.

4.5.2.3 Safety lock. During conduct of the test in 4.5.2.2 the action of the safety lock engagement and release mechanism shall be observed to determine compliance with 3.5.4.3.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Preservation, packaging, and packing for each level of protection shall be in accordance with the applicable requirements of Specification MIL-P-3684 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

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6. NOTES

6.1 Intended use. The paper cutters are intended for use in trimming, squaring and shearing of books, pamphlets and quantities of paper in Military installations where the volume of work does not warrant the use of motor driven machines.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size paper cutter required (see 1.2).
(c) When a first article is required (see 3.1, 4.3, and 6.3).
(d) When split back gage is required for type II cutters (see 3.5.3.3.2.1).
(e) When floor stand is required for type I paper cutters (see 3.7).
(f) Applicable levels of preservation and packaging and packing (see 5.1).

* 6.2.1 Contract data requirements. Any requirements for contract data items, such as equipment manuals for the items covered by this specification should be included in DD Form 1423, Contract Data Requirement List, and cited in the contract.

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Review activity:

Army - ME

Preparing activity:

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Project No. 3610-0183

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-43227A CUTTERS, PAPER, GUILLOTINE, HAND OPERATED

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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